



HPC

Getting started

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Helios is a new computing cluster based on modern CPU and GPU architectures. It is intended for deep-learning and machine-learning and for applications requiring newer Operating System.

Main features

- GPU computing. **Only** for deep-learning and machine-learning applications
- GPU are based on NVIDIA A100 technologies
- Parallel MPI and SMP based applications
- Interactive sessions for post-processing and graphical applications
- Resources management are based on **Slurm**
- High speed network (Intel Omni-Path) 100G/s
- Parallel file system based on BeeGFS
- Software and packages are installed using **Spack** package management tool
- Operating System is Rocky Linux 8.5

HPC workflow

A typical use case for an HPC cluster is as follows:

1. Connect to the cluster
2. Transfer your files to the cluster
3. Compile your code and test it or load installed software using module
4. Create a job script
5. Submit your job
 - i. wait while
 - ii. your job gets into the queue

- iii. your job gets executed
 - iv. your job finishes
6. Move your results

In this documentation, we will learn each of these points.